

3.3.4 Metals

Metals are used for building structures of robots mainly in industry, but some enthusiasts do use them for making their robots because of the sheer strength they provide. You may get any metal you want at junk shop, but it's always better to avoid choosing metal as robot building material if you are a beginner and do not have the luxury of a workshop at your disposal. Most commonly used metal for robot building is aluminium which is light, offers good strength, and is also immune to corrosion.

3.3.5 Motors

A motor is a mechanical device which converts electrical energy into mechanical energy. Any robot, which needs to move, must have at least one motor. If you want your robot to move like a car, attach motor with wheels to your robot. If you want your robot to have a mechanical arm or wrist, put a motor where you want a joint. Anywhere you need motion, motor is there to help.

There are many kinds of motors available, but only some of them are utilised for robotics.

This section will examine the various kinds of motors, how they are used and where they can be used.

There are two broad categories of motor, based on the form of power supply they require: AC motors (works on alternate current) and DC motors (works on direct current).

Mainly, direct current dominates the field of robotics, especially for amateurs and hobbyists. This is because, in robotics, we prefer on-board power source, which can be provided by DC batteries very easily. Mainly industrial robots use motors designed to operate from AC. Some of those robots convert the AC power to DC, and then utilize it, while some directly use AC current, and thus require the use of AC motors.

There are many kinds of DC motors available in the market, and you should buy a DC motor which suits your application the best. For example, you should always buy reversible motors (those which can rotate in both directions) because very few robotic applications call for just unidirectional (one-direction) motors. Generally DC motors are bidirectional, but some design limitations may restrict reversibility. The most prominent factor is the commutator brushes. If the brushes are slanted, the motor probably can't be reversed. Another factor which may prevent the reversibility is the